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OlfacKit: A Toolkit for Integrating Atomization-Based Olfactory Interfaces into Daily Scenarios

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ABSTRACT
To increase the ubiquity and merging of Olfactory Interfaces (OIs) into daily scenarios seamlessly and aesthetically, we propose OlfacKit, a fully integrated pipeline, which consists of (1) *an interactive design tool* that enables novice users to quickly model the physical forms of OIs and automatically generate fabrication-ready designs; (2) *a construction workflow* with a series of assembly kits and a video tutorial; (3) *a WebUI* for rapidly testing and prototyping smell control through Wi-Fi communication. We tested the atomization performance with different voltages and aroma liquids and further guided the implementation of the WebUI. Six applications have been explored, showing the flexibility and scalability of our toolkit. We conducted a workshop to evaluate the feasibility and usability of OlfacKit, involving 15 participants who proposed 108 creative ideas and implemented 13 working prototypes. We reported the detailed findings and discussed the insights and implications for future design.

KEYWORDS
Toolkit design; atomization-based olfactory interfaces; smell experiences; interactive design tools; 3D printing; fabrication

1. Introduction

Compared to other human sensations, olfaction can powerfully affect human emotion (Ehrlichman & Bastone, 1992), behavior (Wilson et al., 2006), cognition (Rouby et al., 2002), and memory (Herz & Engen, 1996; Sullivan et al., 2015) unconsciously. In 2001, Kaye first introduced aroma as a dynamic and promising medium into the HCI community (Kaye, 2001). In the following two decades, a growing number of researchers and designers have developed various olfactory displays for notifications (Dobbelstein et al., 2017), well-being (Amores et al., 2018; Amores & Maes, 2017), immersive experiences (Hu et al., 2021; Nakamoto et al., 2020), synesthesia (e.g., olfactory-gustatory (Narumi et al., 2011; Velasco et al., 2018), and visual-olfactory (Mochizuki et al., 2004)), attention management (Spence et al., 2001), in-car interaction (Dmitrenko et al., 2020), as well as Machine Olfaction (Nakamoto, 2016). However, as an emerging area in HCI, our understanding and exploration of olfaction are still at the embryonic stage. Even previous research has investigated and summarized detailed opportunities for using odors in daily life (Obrist et al., 2014), yet, what are the extended applications, how smell-enhanced technology merges with existing settings, and what would be the execution strategy are still unknown. In addition, it is also essential to consider aesthetics (e.g., shape, color, and material) other than technically and functionally working (Alben, 1996; McCarthy & Wright, 2004), which we believe could facilitate the widespread acceptance and willingness of individuals to use this emerging technology.

To address these challenges, we aim to increase the ubiquity, flexibility, and scalability of OIs by involving more ordinary people with less or without 3D modeling and hardware programming skills. We hope to study what and how they create using our toolkit with their design concerns and further merge OIs into daily scenarios *seamlessly* and *aesthetically* with extended applications (Figure 1).

Therefore, we propose OlfacKit, a fully integrated pipeline, including (1) an interactive design tool for quick modeling the physical form of OIs, (2) construction workflow with a series of assembly kits, and (3) a WebUI supporting fast connection through Wi-Fi communication and rapid scent test and control. Firstly, our interactive design workflow supports end-users to fast model the physical form depending on the on-demand scent release, which includes scent container type and specification, scent release direction, permutation of the multi-scent, arrangement for scent container and control board, and attachment methods. We also support users importing 3d models for decorating the scent containers. Then, we applied a plug-and-play construction method facilitating the device and hardware assembly with a series of materials: standard 3D-printed piezo holders, PTFE tubings, cotton, custom Printed Circuit Boards, odor kits, connecting wires, and a video tutorial. Besides, we offered a WebUI for rapid smell test and control through Wi-Fi communication without accessing professional programming software. The users can modify and simulate the

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Figure 1. This paper proposes OlfacKit, which facilitates novice users to rapidly create, customize, and fabricate atomization-based olfactory interfaces. We demonstrated a series of applications created via OlfacKit. These designs could be lightweight and aesthetic to be worn on the body as a scented button (B) and jewelry (B), attached to daily objects merging with mugs and bottles augmenting drinking behaviors (A), or placed on the desktop for decoration (C), smell training, odor reproduction (E), and IoT communication (F).

smell experiences regarding the scent release distance, desired scent intensity, and working mode of OIs, and remove the odor residues by using cleaning mode for long-term use. Furthermore, we chose ultrasonic atomization for the scent release mechanism and evaluated the performance by applying different voltages and scented liquids. We also validated our method's usability, feasibility, flexibility, and scalability through six demonstrations and a workshop study. We discussed the detailed insights and implications for future design.

In summary, our contributions are listed as follows:

- Implement OlfacKit, including (1) an interactive design tool for quick modeling the physical forms of OIs, (2) a construction workflow with a series of assembly kits, and (3) a WebUI supporting fast connection and rapid scent test and control through Wi-Fi communication. We also open-sourced OlfacKit on GitHub (https://github.com/OlfacKit/Olfactory-Interfaces-Toolkit.git);
- Conduct an evaluation on atomization performance by applying different voltages and odor sources to facilitate and guide the implementation of our WebUI;
- Showcase a series of design examples to prove the flexibility and scalability of OlfacKit;
- Validate OlfacKit through a formal workshop involving 15 participants and further discuss the insights and implications for the future design of Olfactory Interfaces.

2. Related work

2.1. Olfactory display mechanisms

Olfactory Displays (ODs) are the mechanisms generating the scent from resources and delivering a certain amount of odor molecules to a target location (Kaye, 2001; Nakamoto, 2013; Yanagida & Tomono, 2013). Five scent release mechanisms were mainly explored and used, including ultrasonic atomization (Amores & Maes, 2017; Batch et al., 2020), heating (Choi et al., 2011), fan (Dobbelstein et al., 2017), air compressor (Dmitrenko et al., 2017), and Aircanon (Hu et al., 2021; Yanagida et al., 2004).

Atomization releases the odor by generating the scented liquid into a fine mist through ultrasound using Piezoelectric Transducer (Arnau, 2004), which is also commonly used in commercial products such as humidifiers. In the HCI field, prior work has applied this technique for immersive VR experiences (Brooks et al., 2020), stress management (Lin et al., 2020), anaglyph video smell presentation (Abid et al., 2015), odor reproduction (Wen et al., 2019), conveying emotions and information (Batch et al., 2020; Cheok & Karunanayaka, 2018; Patnaik et al., 2019; Xiang et al., 2016) and indoor smell regulation (Wang, Meng, et al., 2020). Other researchers explored the form factors (aesthetics and modular structure) of the atomization-based scent delivery mechanism (Amores & Maes, 2017; Amores et al., 2019; Wang, Amores, et al., 2020). Heating accelerates the movement of odor molecules using liquid and solid odor sources and is always utilized by combining with the fan generating an invisible fragrant airflow. However, uncertain heating parameters for smell release (e.g., delayed odor feedback) could also raise the complexity of precise smell control for end-users. Other methods such as the pumping and valve (Yamada et al., 2006), bubble maker (Seah et al., 2014), and Aircannon (Hu et al., 2021; Yanagida et al., 2004) have also been utilized. Yet, their bulky forms and mechanical noise could pose great challenges in creating natural and seamless designs for extended scenarios within daily settings, e.g., for sleeping, studying, or used as wearables. In addition, even though the electrical stimulation method is lightweight, yet still complex to implement due to the accessibility and social acceptability (Brooks et al., 2021).

Among the above methods, Piezoelectric Transducers greatly benefit our motivation for their tiny, lightweight, soundless, easy-to-control, and commercially accessible properties. Therefore, as a starting point for merging Olfactory Interfaces into daily scenarios naturally and aesthetically, we explored the air-tight and easy-to-assemble mechanism with two types of scent containers and further proposed the permutations for multi-scent releasing and decoration strategy for aesthetics.

2.2. Toolkit design for sense of smell

As a powerful medium for the human body, a few researchers and designers have made efforts to build a toolkit for the sense of smell within different contexts. Leret et al. proposed *Smell Memory Kit*, taking advantage of the motivational, story-sharing effects of scents to evoke and share autobiographical episodic stories among the clients for addiction care (Leret & Visch, 2017). Sissel et al. invested in two scent capsule designs that could be worn as a necklace to create personalized smell memories (Tolaas, 2015). Both designs work non-digitally. In HCI, Saleme et al. proposed a DIY method for individuals to create their multimedia environment, including olfactory and gustatory displays (Saleme et al., 2020). Obrist et al. developed *Owidget*, a GUI for the users to create and replicate the device-independent smell experiences (Maggioni et al., 2019). Most recently, Lei et al. proposed a rapid design and fabrication method using low-cost cardboard for creating olfactory interfaces (Lei et al., 2022).

Compared to this work of using disposable cardboard creating rough paper-based prototypes (Lei et al., 2022), we aim to merge the Olfactory Interfaces into daily scenarios seamlessly and aesthetically by proposing a fully integrated pipeline. We utilized 3D printing to construct the physical form of OIs, considering its flexibility and efficiency for creating functional, durable, and aesthetic (e.g., shape and color) artifacts. We intend to follow the established knowledge and the wisdom of Personal Fabrication that highlights the computing of physical properties through parametric modeling and generative design (Baudisch & Mueller, 2017; Koyama et al., 2015). For instance, *RetroFab* empowers non-experts to retrofit physical interfaces by 3D printing the automatically generated enclosure structure (Ramakers et al., 2016). *Reprise* helps users specify, generate, and customize 3D printable adaptations on everyday objects (Chen et al., 2016). Similarly, we provide a design toolkit to help novice users construct 3D-printed scent delivery devices. Additionally, we also customized a control board with a WebUI to enable users without programming skills to rapidly build their personalized olfactory experiences.

3. Design criteria

3.1. Flexibility and scalability

It is crucial to support the users in *customizing diverse scent releases* within various and extended daily scenarios. Hence, we offer the scent delivery manipulation in terms of the (1) *type of odor sources*, (2) *amount of liquid aroma*, (3) *number of odors*, (4) *scent release direction*, (5) *distance*, (6) *intensity*, and (7) *frequency*. To enlarge the odor source pool, we provided four types of commercial scented liquids for the first design parameter, including *oil-soluble essential oils*, *perfume oils*, *water-soluble hydrosols*, and *edible essences*. Secondly, to further increase the flexibility in liquid amount and odor numbers, we propose the flexible scent container mechanism by choosing the tubing solution instead of using rigid and custom-made glass bottles. We utilized the PTFE tubes that have been explored previously (Wang, Amores, et al., 2020). We further improved the mechanism into two types, vertical and horizontal, to ensure the containers are flexible in various physical spaces. Besides, to grant the scent release direction with a broad range of angles with a simple mechanism, we use a spherical joint that can rotate 360 degrees with a range of 270 degrees. We further developed a customized control board for controlling four atomizers regarding the scent release distance, intensity, and frequency.

3.1.1. Arrangement and modularity

Individuals are flexible in arranging the scent containers and control board with proposed two solutions: *One Whole Part* or *Two Separated Parts*. The former method could benefit the printing set and assembly, while the latter can be more flexible to adapt to more scenarios (e.g., worn as a necklace (Amores & Maes, 2017). Furthermore, we also support the modular construction of scent containers, PCB, and attachments by using a standard connecting structure.

3.2. Placements and attachments

To merge Olfactory Interfaces with existing objects or be devised as a standalone device, we explored eight typically used attachment methods in four categories, including structural objects, flat surfaces (e.g., wall and desktop), and soft substrates(e.g., cloth and fabrics). (1) The *Clip* method includes the cylinder, rectangular items, and a hair clip for soft materials. (2) *Hook and Ring* method mainly involves the soft and thin substrates; (3) *Sticker and Magnet* are used for a flat surface. (4) *Standalone* method supports the design that could be stably placed on a horizontal surface (e.g., on the desktop).

3.3. Aesthetics

To maximize the aesthetics of Olfactory Interfaces and increase the willingness to use them in daily life, we support users importing delicate, complex, and fashionable 3d models to combine them with the scent release mechanism for decoration. For the horizontal scent container, we support the modification of tubing's appearance (straight or bend), e.g., shaping like a flower. In addition, we also offer additional materials to maximize user preference, such as jewelry findings, paint, and sewing kits, which are commonly used within the DIY and Maker's community.

3.4. Safety and sensitivity

3.4.1. Air-tight and liquid-proofing

When designing the atomization-based olfactory interfaces using liquid aroma, confirming an air-tight scent delivery mechanism is vital in case of any scent lingering or leaking that causes electrical short circuits. Hence, we devised the scent container using silicone plugs at both ends of the tube and seal rings between the piezo holder and the 3D-printed bottom part (design details are shown in section 4.1.1). We offer a board case for protection because exposed circuit boards can also pose dangers (e.g., insulation and leaking liquid). Besides, due to the varying sensitivity of the users, we recommend *using highly-diluted scented materials* in case of allergy or habituation, especially highly-concentrated aromas, such as essential oils and perfume oils (Figure 2).

4. Interactive design tool

This section presents the details of the primary mechanisms, adjustable parameters, arrangement logic, workflow, and software implementation of our design tool. All the standard models are provided in the supplemental materials.

4.1. Scent container customization

The piezo's shape and atomization principle (contact with the liquid or soaked cotton directly) lead it to a natural cover for the scent reservoir. Iterated and improved from previous work. (Wang, Amores, et al., 2020; Amores & Maes, 2017), we proposed two types of scent container designs, vertical and horizontal, using 16 mm and 10 mm piezos, respectively.

4.1.1. Tubular scent container

We modulize the scent container of the piezo-based olfactory display to simplify the assembly process. We build two standardized piezo holders for the 10 mm and 16 mm piezo with a bayonet mechanism (Figure 3(a–2)) that can tightly fasten the piezo to the tube with a simple" press and rotate" operation. We utilize a hollow-carved design (Figure 3(a–4)) to display the scent volume for easy observation of the vertical scent container. To avoid the device being too high, we have the horizontal design of the scent container, which distributes all the PTFE tubes horizontally. To ensure the feasibility of this approach, we add limiters and a pedestal to hold the tubing in the desired shape. We also ensure that one limiter is placed at the end of each tubing (Figure 2(A)-Horizontal Design & IoT application in Figure 1(F)). Moreover, we also support the users choosing different attachments for scent containers and control board cases when selecting the" Two Separated Parts" arrangement. To ensure enough space for generating the guiding, we define different placements for two containers (Figure 3(a–10)), which will significantly influence the scent release direction.

4.1.2. Amount of odor source

Since the capacity of the odor resource mainly determines the size of the scent containers, we support the users to customize the desired aroma volume to adapt to various usage scenarios. Figure 3(b) illustrates the volume selection plan, and the

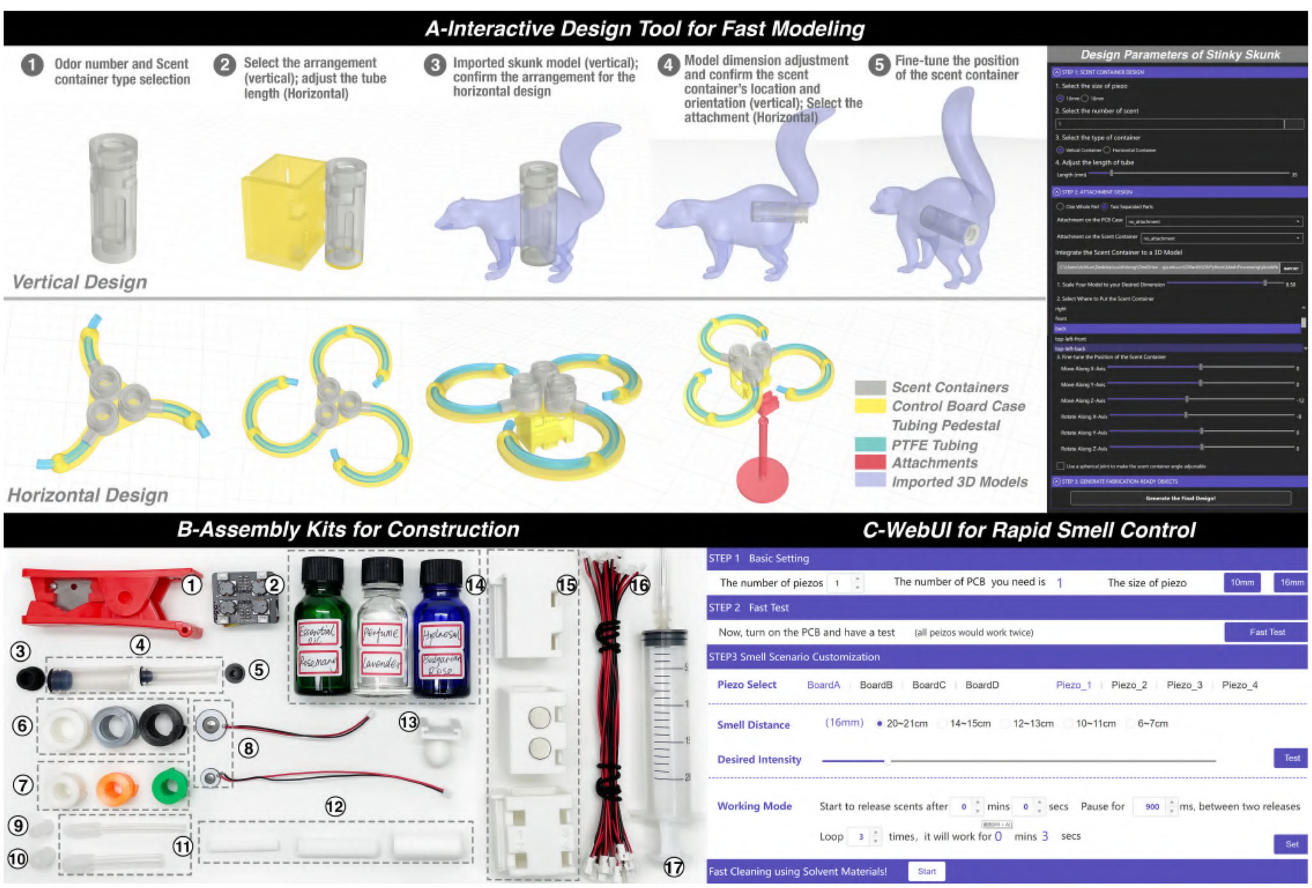

Figure 2. OlfacKit overview. (A) an interactive design tool that enables users to fast model the physical forms of OIs and automatically generate fabrication-ready designs. The right image shows the main design parameters of our skunk demo. (B) the construction kits with a series of assembly materials. (C) a web-based user interface for rapid testing and prototyping smell control through Wi-Fi communication.

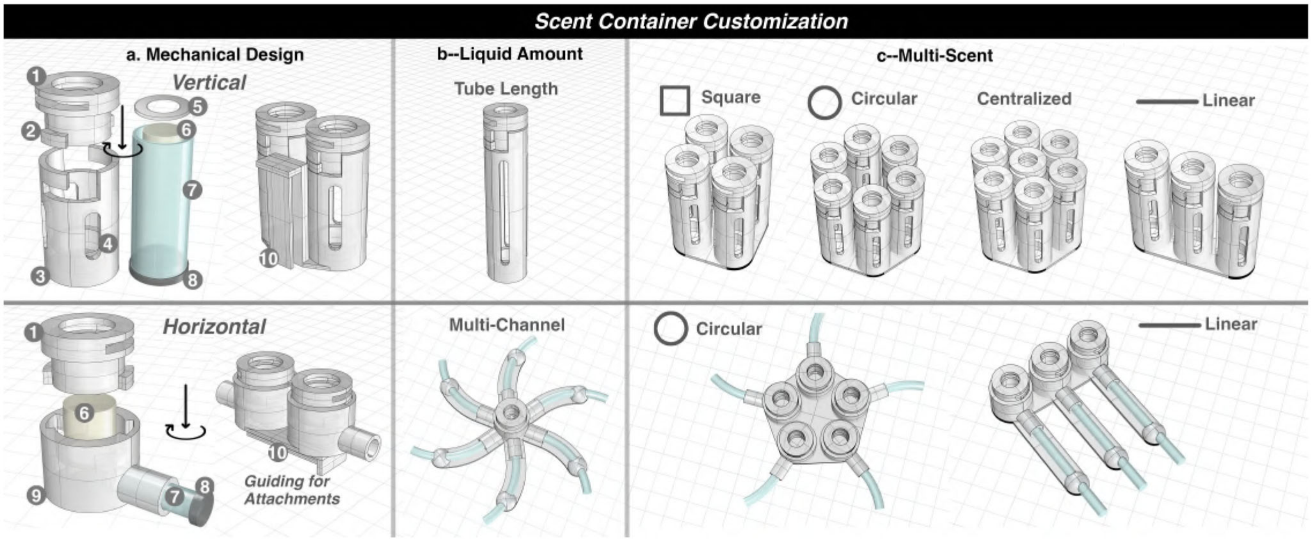

Figure 3. Scent container customization. Image a: basic mechanical construction for vertical and horizontal design, including (1) the standard piezo holder, (2) a bayonet mechanism; (3) the bottom part (vertical type); (4) a hollow-carved design; (5) silicone seals placing under the piezo; (6) cotton; (7) PTFE tubing; (8) air-tight silicone plugs; (9) the horizontal bottom part. Image b: solutions for modifying the liquid volume. Image c: permutations for multi-scent containers.

corresponding generated forms for the vertical and horizontal design. Notably, due to the tubing solution (thin form factor, liquid gravity, and placement) for the scent container design, we recommend using a tube length less than 200 mm. Nevertheless, even with the length limitation, we can also have a considerable capacity for the odor resources within scenarios, for instance, approximately 4.5 mL with a 60 mm length tubing (Diameter = 16 mm) for one scent capsule of the smell training device (Figure 1(E)) and 2 mL for the small ring design (Diameter = 10 mm) using the multi-channel solution for one scent (Figure 1(B)).

4.1.3. Multi-scent releasing

We offer a series of permutation designs when users have the requirements of using multiple odors, including some regular graphic patterns and the distance modification between each piezo (Figure 3(c)). We default the distance among piezos to 2 mm as the minimum threshold to avoid broken models or complex assembly from 3D printing errors. By adjusting the distance between each piezo, the user can have either an outlined (e.g., round, triangular) or a space-filling permutation.

4.1.4. Adjustable scent release direction

We build a standard spherical joint mechanism consisting of a ball and socket allowing rotational movements (Figure 4). The scent container can be easily fixed in a specific direction by the pre-set 3D printing tolerance to increase friction. Since the direction is flexible enough under certain scenarios (e.g., wearable), we define this structure as optional.

4.2. Control module

4.2.1. Hardware design

To simplify the hardware connection and maximize the capability of the scent control, we customized a minimized print circuit board (PCB) that can control four piezos. The control module consists of a motherboard and a driver board (Figure 5(A,B)).

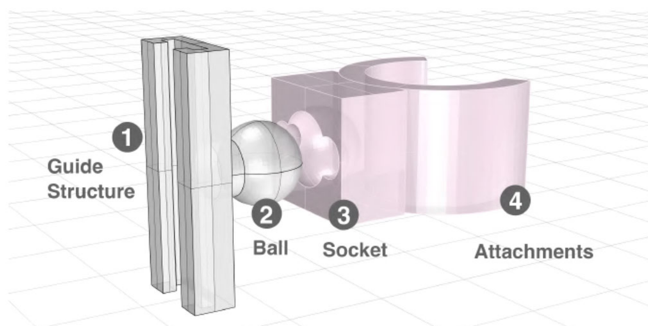

Figure 4. The design of the spherical joint mechanism.

4.2.1.1. Motherboard. The motherboard can communicate with cloud platforms through Wi-Fi with an ESP32-based microcontroller (ESP322-PICO-D4), which provides a 2.4 G Wi-Fi patch antenna (CrossAir CA-C03). The microcontroller can be programmed using the Arduino or MicroPython (MicroPython, 2022) development environment. Two sets of I/O ports are reserved to support simultaneous access to two different sensors. The motherboard can be charged or updated by the control program through Type-C. Advanced users can quickly develop desired functions they want based on our examples.

4.2.1.4. Driver board. Standard piezos of different sizes (10 mm, 16 mm) require a high voltage (65 V) with two different frequencies (165 kHZ, 110 kHZ) to convert liquids into a mist. The boost circuit on the driver board can amplify voltage even if the electrical signals from the motherboard have different frequencies. The driver board uses similar circuits to ensure four piezos work simultaneously. This architecture makes piezos to be hot-swappable.

A Li-ion battery (3.7 V, 300 mAh) powers the whole system, which allows for 3.5 h of the system running if four piezos (16 mm) are working 20 ms per second (2.6 h, 10 mm) and the device remains online through Wi-Fi. The weight of the whole control board is 17 g.

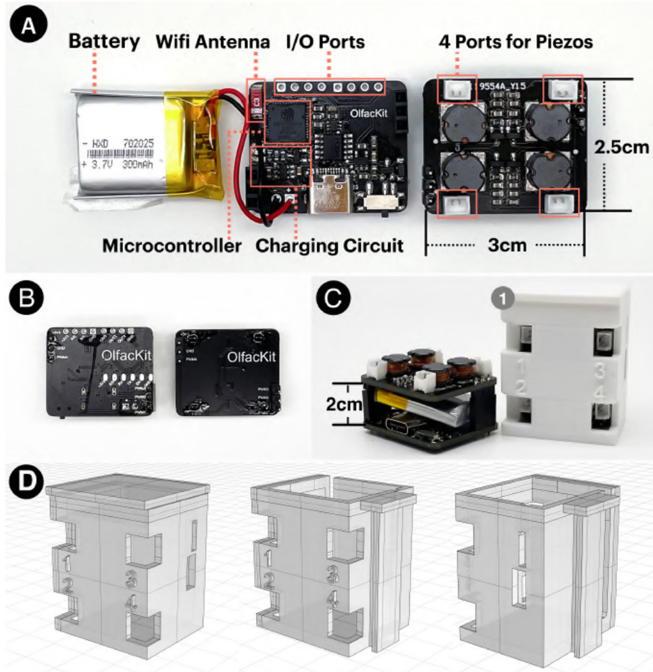

Figure 5. The custom Printed Circuit Board for smell control. (A) The top view of the control board, (B) the bottom, and (C) the stack-up view of the control board. C-1 shows the standard print of the board case. (D) shows the 3d model of the board case design, including the standard model (left) and guiding structure at the right (middle) and back (right) sides of the board case.

4.2.2. Board case design

We build a standard model of the board case by reserving space for four piezos, a switcher, a Type-C port, and sensors (Figure 5(C)). We also label the piezos on the case from 1 to 4 to assist users with scent control. The board case plays a crucial role in connecting the spherical joint and attachments. We define two sides (right and back side) of the case for connecting with a standard guide structure considering safety use (Figure 5(D)). We aim to enable users to quickly replace different attachments for different scenarios.

4.3. Attachment

We propose eight attachment methods for adapting various physical scenarios (Figure 6). The user can adjust their parameters according to personal preferences and measurements (e.g., the diameter of a mug). The attachments can be connected directly with the spherical joint, the PCB case, or the scent container. Our design tool generates the attachment with a socket for the direction adjustment requirement, or a guiding structure when no spherical joint is required. In addition, since the attachment of the "hair-clip guide" and "ring" are usually small and combined with soft substrates, e.g., the band and the collar, we default that these two methods can not be coupled to the spherical joint.

4.4. Arrangement

We support users in selecting the arrangement of the scent container and the control board for both vertical and horizontal designs, including "One Whole Part" and "Two Separated Parts." In the first condition, the scent container and the board case will be printed as a whole set, and the selected attachment is matched automatically to the board case (Figure 1 A mug design). Nevertheless, when generated with the board case together, the large-angle overhang of the horizontal design becomes challenging for 3D printing with a consumer-level FDM printer. Therefore, we print the scent container and board case individually and use magnets to stick them together as a whole part (Figure 13). In the second condition, we support users choosing desired attachments separately for the scent container and the board case.

4.5. Design workflow

Figure 2(A) shows the user interface of our design workflow. Overall, individuals can follow the non-linear design process to adjust the parameters and preview fabrication-ready designs at any stage. Our workflow consists of the following four steps. Once satisfied, users can export all the geometry as 3D printable STL files.

4.5.1. Step 1: Scent container customization

Users start by modifying the appearance of the desired scent container, including the size of the piezo, the number of controlled odors, the type of scent container, and the length of the PTFE tubing that indicates the liquid amount of the scent container (Figure 3(a–7)). Users can choose straight or bent tubing for aesthetics if a horizontal container is selected (Figure 3(c)). Additionally, the system provides several dynamic permutations of multiple scent containers in both vertical and horizontal designs for users to choose (Figure 3(c)).

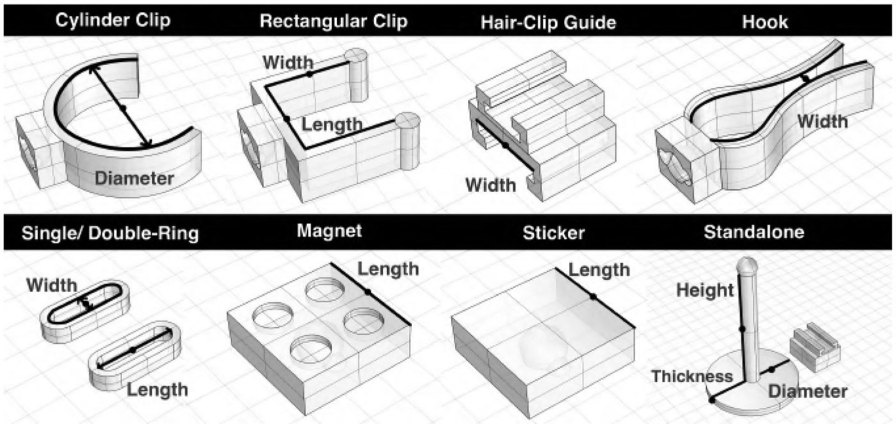

Figure 6. This figure shows our eight attachment methods and their adjustable parameters.

4.5.2. Step 2: Arrangement selection

Next, for adapting to variable physical scenarios, our system provides two arrangement methods, i.e., using the scent container and control board as a whole part, or separately (Figure 2(A–2)). Once the "One Whole Part" selection is confirmed, the software will generate a new design and simultaneously enable the users to select the placement on the board case for attachment. The guiding structure can assist users in visualizing the spatial position of the attachment (Figure 3(a–10)). While the other situation is selected, users could choose desired attachments separately.

4.5.3. Step 3: Attachment selection

Before selecting a specific attachment, users are optional to keep the spherical joint. The software automatically generates the matched guiding structure (Figure 3(a–10)), allowing users to concentrate on the attachment selection and how to use them. Users can also adjust the parameters of the attachment based on desired preference or pre-measurement.

4.5.4. Step 4: Decoration with arbitrary 3D models

Users can further decorate the container by combining it with additional 3D models, which could be downloaded online or designed by themselves. After the users import the prepared 3D model (STL or 3DM file format), we align the model with the scent container by their volume center, helping the user perceive their relative size (Figure 2(A–3)). Users can also scale the imported model with a slider. Next, users are allowed to choose from 14 directions (i.e., from the center of a cube to its six faces and eight corners) to adjust the spatial angle between the scent container and the imported model (Figure 2(A–4)). During this process, the system automatically moves the top of the scent container to the outermost surface of the imported model. Besides, the user can further move or rotate the scent container along the X, Y, and Z axis for fine adjustments (Figure 2(A–5)). To ensure the functionality of the scent container, the system automatically performs boolean operations between it and the imported model. Note that this step is only for users choosing the vertical design and the" Two Separated Parts" arrangement.

4.6. Software implementation

We create our interactive design tool within Grasshopper (Rhinoceros, 2022a), a visual programming language and environment that runs within Rhino. We implement all the geometry processing functions using RhinoPython (Rhinoceros, 2022b) and RhinoCommon API (Andheum, 2023). We also use Human UI (Andheum, 2022), a plugin for Grasshopper, to develop our user interface. Everything runs on an Alienware computer with a Windows system.

4.6.1. Permutations of multiple scent containers

To generate the different permutations of multiple scent containers, we first construct a series of translation vectors to move a single container geometry to new placements (Figure 3(c)). We then divide the edge curve of every container's bottom surface into 500 segments and use the convex hull of these points to generate the pedestal surface. We extrude the pedestal surface along its normal vector and perform a boolean union operation between it and the scent containers.

4.6.2. Pedestal generation for holding horizontal tubings

When users choose the bent tubing while customizing the horizontal scent container, we constrain the minimum radius of curvature based on the pre-tested tubing's ability to bend. Therefore, based on the customized tubing length, we can calculate the desired arc and use it to generate the tube geometry by creating a pipe surface. The arc is also offset to generate the bent tubing pedestal, which is joined with tube limiters to hold the PTFE tubing (Figure 3(b)). To prevent the intersection of multiple tubing pedestals when the tubing length is increasing, we use a collision detection algorithm that recursively increases the radius of curvature by 0.1 mm each time for minimizing the area.

4.6.3. Combine the scent container with 3D models

When the user imports a 3D model and adjusts the spatial relationship between it and the scent container, we ensure the functionality and manufacturability of the generated designs by the following two strategies, similar as the method in RoamFab (Luo et al., 2022). (1) *The scent container intersects the imported model*: To simplify the assembly process after 3D printing, the system always moves the scent container to the outermost surface of the imported model during the user adjustment process (Figure 2(A–4&5)). According to the scent release direction, we construct a line from the top center of the container and find its intersection points with the imported model. Then we use the farthest point to construct a vector and move the container. No movement will be made when there is no intersection point, meaning the scent container could be outside the imported model. Next, the system performs boolean operations (Mesh or Brep geometry) on the imported model to make room for the scent container and the top area for odor release. (2) *The scent container does not intersect the imported model*: The system will automatically generate a cylinder connector (Koyama et al., 2015) to combine them, as shown in Figure 14(D).

5. Construction

OlfacKit comprises a series of materials and kits facilitating individuals to assemble the device rapidly (Figure 2(B)). We detail the assembly kits and workflow in this section.

5.1. Assembly kits

5.1.1. Standard materials

(1) The piezo holders (Figure 3(a–1)) for both 16 mm and 10 mm piezo (Figure 2(B–8)) are pre-printed due to the modular design. We also prepare other standard prints, such as the spherical joint (Figure 2(B–13)) and the board case (Figure 2(B–15)). Nevertheless, individuals are also flexible in using existing prints or re-printing them with desired colors. (2) To construct the scent container, we use the PTFE tubings (Figure 2(B–4 and 11)), cotton (Figure 2(B–12)), air-tight silicone plugs (Figure 2(B–3, 5, 9, and 10)), and silicone gasket. Different material specifications are chosen for implementing the vertical and horizontal design. We provide a detailed datasheet for assembly, which was attached in the supporting files. (3) Multiple control boards (Figure 2(B–2)) are prepared for scent control. To ease the hardware connection for adapting to various applications, we also provide the extension wires with terminal blocks (Figure 2(B–16)).

5.1.2. Odor sources

To facilitate odor selection and stimulate the smell experience, we prepared 45 different odors, which could be categorized into four types of liquids, including essential oils (7), perfume oils (28), hydrosols (3), and edible scents (7). The former two oil-based odors used food-grade Ethanol 190 Proof (95%) as the dilution substrate, while the latter used pure water. Besides, the types of fragrances included flowers (e.g., lavender, rose, gardenia), fruits (e.g., litchi, banana, orange), wood (e.g., sandalwood, pine, cypress), and scents from daily life (e.g., coffee, ocean breeze, grass, sunshine). We also prepared diluted coffee (1%, 3%, 5%, 10%, and 20%) to assist the users in choosing their desired intensity and avoiding allergy and habituation when using high-intensity raw scented materials. We selected these ratios based on online instructions and precautions for using scents.

5.1.3. Additional kits

We also provide additional kits commonly used within DIY and Maker communities, such as sewing kits, jewelry findings, and paints, to maximize user preference.

5.2. Assembly workflow

Figure 7 shows the main process of the device construction. Our toolkit provides a video tutorial in the supplemental documentation to support users visually. We summarize this process in two steps:

5.2.1. Step 1: Materials preparation

Before the device assembly, users need to prepare all the materials with proper specifications based on our datasheet. We provided users with fabrication tools for customizing the length of the tubes and the cotton using the tubing cutter (Figure 2(B-1)) and scissors.

5.2.2. Step 2: Device assembly

The order we recommend for assembly is from scent container, spherical joint, attachment, electronics, and finally, liquid filling. The assembly of the scent container follows the steps of inserting the air-tight plug and then the cotton. Next, the piezo and the control board are inserted into the 3D-printed piezo holder and case correspondingly. To protect electronics from liquids, we recommend filling the scented materials at last. Notably, due to the mechanism of the horizontal design, we suggest filling the liquids with the device placed horizontally before inserting the plugs. Finally, plug in all the piezos to the control board and ensure all the components are assembled tightly.

6. WebUI for rapid smell control

6.1. Atomization performance

The smell experience highly correlates with the scent delivery distance and intensity. Therefore, to precisely control the scent release and facilitate the implementation of WebUI, we conducted two experiments to comprehend the atomization performance (burst height and atomized odor quantity) when applied with different *voltages*, *aroma liquids*, and *concentrations*.

We utilized the atomizers with a diameter of 10 mm (1050 micro-pores, diameter = 10 μm) and 16 mm (800 micro-pores,

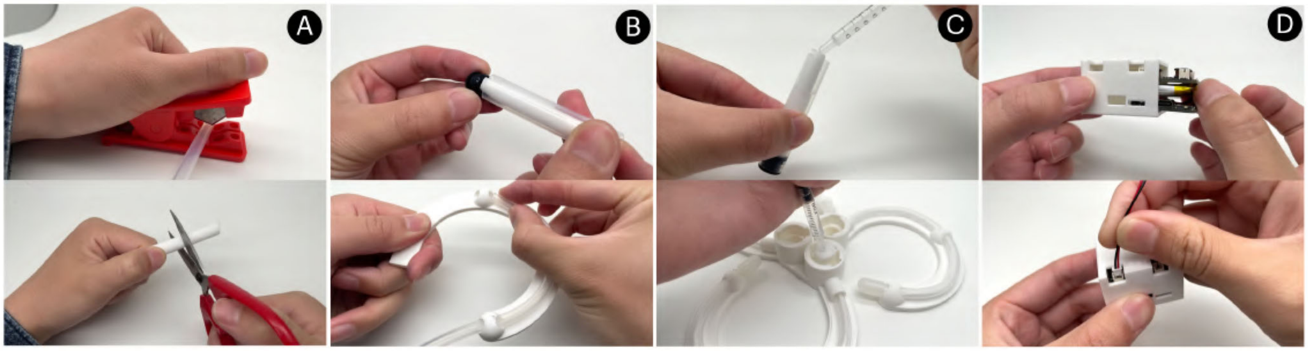

Figure 7. The figure shows the main steps for assembling the scent delivery device for both the vertical and horizontal design, including (A) materials preparation, (B) tubing assembly, (C) liquid aroma filling, and D) electronics assembly.

diameter = 13 µm), and the smell training demo (Figure 1(E)) for atomization. We chose the food-grade Ethanol 190 Proof (95%) to dilute the essential oils and perfume oils, while using pure water to dilute the edible essence. For each test, the device was covered with a transparent box ($50mm \times 50mm \times 60mm$) to ease the observation and reduce interference from the wind. We set the atomization mode of the piezos as working 100 ms, pausing 900 ms, and last 400 times for one round. We kept testing this four times to avoid errors. We measured the corresponding burst's height (from piezo to the end of the mist) and recorded the weight of atomized liquids (before and after atomization using a scale). Finally, we calculated the averages and reported the final results below along with Figures 8 and 9.

6.1.1. Performance with different voltages

Firstly, we found that even though the voltage was positively related to atomization performance, the results were not linear. The reason might be the handcrafted cotton and the changing tightness between the piezo and cotton. Nevertheless, we could also find the results separated in several parts; for example, for 16 m*m* piezo, the burst height and atomized quantity changed obviously from 10% to 30% voltage (e.g., average height is 0, 6.6 cm and 10.2 cm correspondingly), while relatively similar from 40% to 60% (height is 12 cm, 12.2 cm, 12.7 cm), and increased vastly again from 80% to 100% (height is 13.3 cm, 17.2 cm, 19.8 cm). This could also be observed by 10 mm piezo but less obviously. To further implement WebUI, we divided the burst height and voltage ratio into five categories: 20–21 cm (100%), 14–15 cm (80%), 12–13 cm (70%), 10–11 cm (30%), and 6–7 cm (20%) for 16 mm atomizer; and 12–14 cm (100%), 9–10 cm (80%), 6–8 cm (70%), 2–3 cm (60%), and 1 cm (50%) for 10 mm piezo.

Secondly, we observed the burst's visual effect. We found it also matched the rules of the atomization performance with different voltage ratios, from "like a yarn" (20% for 16 mm piezo and 50% for 10 mm piezo), and barely can see to get thick and ultimately powerful. Additionally, during the test, we also calculated the number of bursts that a single scent container (Tube length = 60 mm, Liquid volume

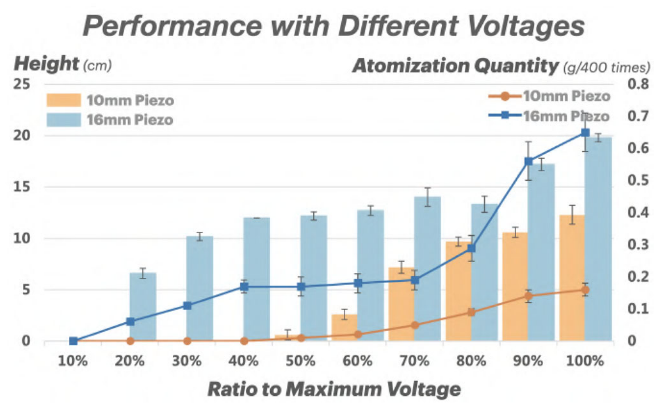

Figure 8. In this test, we utilized 10mm and 16mm piezos to atomize coffee perfume oils (5% diluted) by applying ten different voltage ratios. The maximum voltage of the piezos is AC 10V (20Vpp). We found 16mm piezo showed a vastly higher shot and larger atomization quantity than 10mm for all the voltage ratios, for instance, the 16mm peizo started to atomize at 20% (Height, $M = 6.6$, $SD = 0.49$; Quantity, $M = 0.06$, $SD = 0$), while 10mm piezo needed 50% (Height, $M = 0.6$, $SD = 0.49$; Quantity, $M = 0.01$, $SD = 0$).

= 4.1 mL) could provide, which was approximate 1200 times, and with the mist got weak from 1000 times.

6.1.2. Performance with various odors and dilutions

We found the concentrations might potentially affect the atomized odor quantity for some of the odor materials. However, due to the limited number of testing samples, we can not draw any conclusions about the specific odors and liquid types. Further research could address this issue and provide a detailed scent datasheet. Nevertheless, we could also manipulate and customize the smell intensity by adding the atomized time (ms) adjustment, for example, chose the target scent release distance and adjusting the atomized working time.

Besides, we also found that after thousands of atomizations, the testing model started to get sticky (mainly when we tested the 10% and 20% concentrations) due to the scent release direction (to the top, and the aromatic particles fell on the device). The micropores of the atomizers could also get stuck after long-term use. Therefore, we added a "*Cleaning Mode*" to the WebUI for users' convenience using dilution substrates (e.g., pure water or 95% Ethanol). We set

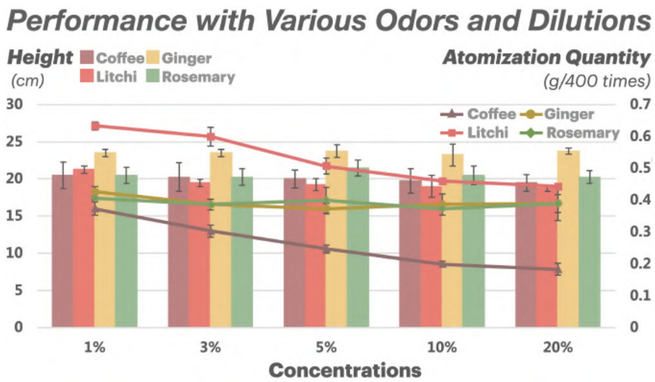

Figure 9. In this experiment, we tested three types of odors, including essential oil (rosemary), perfume oil (coffee and ginger), and water-soluble edible essence (litchi). The voltage we applied for all the dilutions was 100%. We found the liquid aroma has very close values in burst height with different dilution ratios. While in atomization quantity, coffee and litchi showed a decreased value with more concentrated dilutions, and the other two were similar.

the parameters as working 100 ms, pausing 900 ms, and looping 200 times.

6.2. WebUI design and workflow

Our user interface (Figure 2(C)) is built on Aliyun IoT Studio (2022).

6.2.1. Step 1: Basic settings

The user starts by entering the number of piezos to inform the system how many control boards are needed. Then the boards will automatically connect to Aliyun through Wi-Fi with the MQTT protocol.

6.2.2. Step 2: Fast test

Next, the user should select the diameter of the piezos and could perform a 200 ms quick atomization test of each piezo to ensure the scent container works properly.

6.2.3. Step 3: Smell scenario customization

In addition, users can customize their smell experiences on-demand regarding the *scent delivery distance* and the *desired intensity* for each piezo. Our WebUI offers five different distance settings (based on the burst height from the previous test in Section 6.1) under each piezo size, and the intensity can be adjusted between 1 and 300 ms. Besides, we define the *Working Mode* of a specific piezo by the logic "*This scent will be released after A seconds, Pause for B seconds between two releases, and Loop C times for D seconds.*" *A* could take any value from 1 to 3660 s, *B* could take any value from 0 to 100 s, *C* could take any value from 1 to 1000 times, and *D* is calculated in real-time based on the preceding parameters. Through WebUI, the users can create, modify, and simulate dynamic smell experiences regarding the scent delivery distance, desired scent intensity, multiple scent-releasing, and the working mode adjusting scent-releasing frequency, duration, and looping patterns.

6.2.4. Step 4: Cleaning mode

Lastly, the user can choose the Cleaning Mode to remove odor residues on the atomizers using the dilution substrate. Our WebUI currently supports the simultaneous control of four control boards, meaning that it can support 16 piezos separately.

7. Demonstrations

7.1. Stinky skunk!

We envision that olfactory interfaces can benefit education and learning, for example, by simulating how skunks produce, store, and release smelly odors. The robust scent of skunk comes from anal glands inside the rectum at the base of the tail. Therefore, we merged the vertical scent container with a skunk model and set the piezo holder at the corresponding location shown in Figure 10. Meanwhile, since the skunk can emit the spray for quite a long reach, we could use a high-intensity odor resource and set this design with the longest smell distance. Besides, the intense smell created by the skunk is rigorous toxic; to simulate the unpleasant smell, we should choose the stinking odor but not harmful to the human, for instance, the stink or prank perfume (usually extracted from plant fermentation) from the online shop.

7.2. Smell-enhanced ornament

We designed and implemented the scent delivery device merging with aesthetic 3D models as the daily ornament for decoration (Figure 1(C)). We separated the scent-releasing parts and the control board and set the containers in the middle of the rose, acting as the stamens, while hiding the board into the vase. Unlike the other designs releasing a directional scent, we set the height of the aromatic burst very short (3 cm), simulating a natural flower diffusing the scent to create an ambient smell experience and involving pro-active smell behaviors.

7.3. Aromatic wearables

Perfume and jewelry are commonly worn for beauty in daily life. We believe the jewelry (e.g., necklace and ring) has a great potential to be the aroma carrier considering its form factor (e.g., diamond-like), creating a more natural and aesthetic olfactory display. To minimize the dimension, we separated the scent container and the control board module

Figure 10. Front and back view of the stinky skunk.

(Figure 1(B)). We chose the horizontal design (10 mm piezo) and single-ring attachment (Figure 11(A)) for the scent delivery device to be worn on the fingers and a double-ring structure for the control board worn on the wrist with a magic band. The multi-channel design help contain more liquid. We also implemented a scented button (Figure 11(B)) merging with the cloth. We envision that wearers can perfume themselves in need, use them as a scented notification device, or refresh themselves with functional aromas (e.g., lemon and peppermint).

7.4. Flavor-augmentation

We envision that our scent generation module could be merged into the everyday diet, for example, attached to the mug, bottles, bowl, or even spoon for those who suffer from anorexia, taste loss, or decreased appetite. In Figure 1(A), we placed the scent delivery device, including two scent containers (10 mm piezo and 5 ml fragrant) and the control board on the side of the mug through a cylinder clip attachment (Figure 12). We pre-measured the clip's diameter (D = 80 mm) using a caliper. Besides, a spherical joint is applied for the users to adjust the scent release direction. We could also set two single containers aside from the bottles. We recommend using edible aroma and lower scent intensity (10 ms per second) in case of allergy, irritation, and discomfort due to the shorter distance to the eyes.

7.5. Smell training & odor reproduction equipment

OlfacKit can create the design when many scents and real-time feedback are needed (Figure 1(E)), e.g., smell training and odor reproduction in Machine Olfaction. In this design, 16 vertical containers (height = 60 mm) were created with 4.5 mL scented liquid and controlled by four control boards. We chose no attachment for this design to place it on the table. This device may be helpful for olfactory skills training (e.g., perfumers and winemakers) and rehabilitation (e.g., Olfaction Recovery Training in Post-COVID-19) at home. We chose a uniform color (orange) for this design to prevent the bias of the trained people. They can easily replace the scent capsules based on their training schedule. Furthermore, the users (e.g., perfumers and winemakers) can potentially test their recipe by mixing different scents

Figure 11. The design details of the aromatic ring and button that could be worn on the body.

Figure 12. The construction of the flavor-augmented mug.

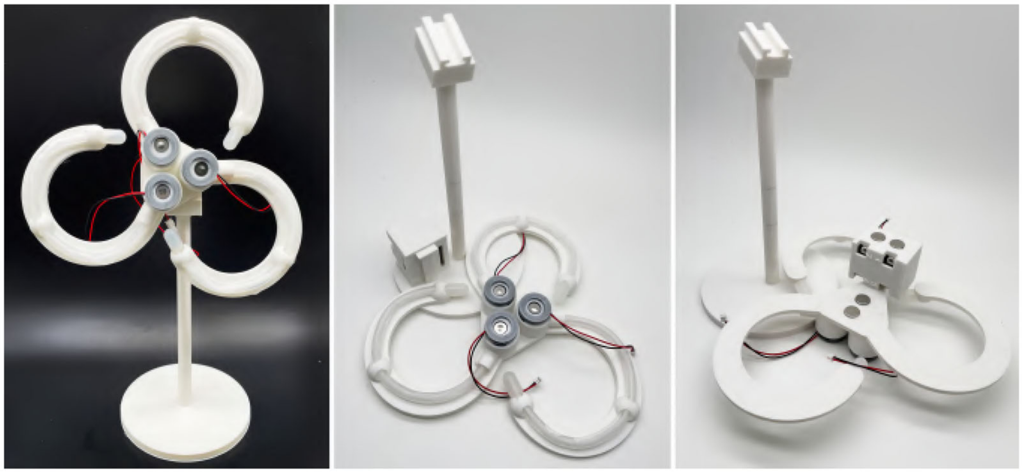
Figure 13. The design of a context-aware IoT device that can release three different scents.

with various intensities. In addition, for odor reproduction, many more scent containers could be created, paired, and communicated together to produce a more precise odor representation.

7.6. Fragrant IoT device

We also envisioned that the scent release could merge with the IoT devices, creating context-aware interactions through sound in daily life (Figure 1(F)). We designed the desktop scent delivery device to release three different odors based on the ambient sound of typing, talking, and knocking, indicating three contexts: working, communication, and notification. All the data is communicated through Wi-Fi, and no physical sensors are needed for this application. We chose the curved design for shaping the tubes like a clover. Furthermore, we proposed a magnet connecting method for the horizontal container to facilitate 3D printing using a consumer-grade FDM printer (Figure 13).

8. Evaluation workshop

The workshop's objective is to evaluate the usability, flexibility, and scalability of OlfacKit and further understand its limitations for future Olfactory Interface design.

8.1. Participant demographics

We recruited 15 participants (7 female, average age 26.53 years, $SD = 4.29$) by excluding those with anosmia or odor allergies, including three college faculties, two undergraduates, and ten postgraduates. With a pre-workshop survey released online, the participants showed diversity in majors, ranging from Industrial Design, Computer Science, Law, Human-Computer Interaction, Public Medicine, Visual Communication to Digital Art and Design. All the participants ranked their skills on 3D modeling using CAD tools ($M = 2.67$, $SD = 1.40$), hardware programming ($M = 1.87$, $SD = 0.81$) and crafting ($M = 3.60$, $SD = 1.14$) in a 5-point Likert Scale (1 = No experience, 5 = Very skilled). Besides, none of the participants reported having experience with designing olfactory displays.

8.2. Procedure and apparatus

The workshop was conducted in a professional laboratory with standard fab shop supplies (e.g., measuring tools and cutters). We also offered a consumer-grade FDM 3D printer and an SLA printer. To budget time for the participants, we split the workshop into two sessions: (1) practicing, brainstorming, and modeling on Day 1 and (2) assembly, scent-releasing control on Day 2.

1. *Introduction & Practice (30 min)*: After signing the consent form, each participant received a detailed workflow manual for documenting their ideas, design considerations, and choices. Then, the host briefly introduced the Olfactory Interfaces and OlfacKit, including the interactive modeling tool, construction kits, webUI, demonstrations, and additional prints with different attachments. Three experimenters explained, in turn, how we designed, assembled, and controlled the skunk demo (Figure 10) as an example. The participants were free to test the prototypes and practice the software with the instructions of the experimenters.
2. *Brainstorming Ideas (35 min)*: We prepared a series of objects as inspirations for the participants to facilitate the brainstorming, such as photos, toys, bowls, speakers, and books. The participants were encouraged to think aloud, record their concepts (∼15 min), and quickly share their ideas by following the host's lead (∼10 min). Finally, the participants were required to clarify and refine a usage scenario to implement in group or solo (∼10 min).
3. *Modeling Design*: After the ideation, the participants started to model the physical form of the scent-delivery devices. They were also free to download the 3D models for constructing and strengthening their concepts. We observed this process, recorded their modeling time, and gave instructions when they were confused.

4. *Assembly & Scent Control (Day 2):* The Participants returned to our laboratory to retrieve their 3D printing items. They were instructed with a video tutorial on the assembly workflow and methods. We also provided acrylic color, spray paint, jewelry findings, and sewing kits that makers commonly use. After the participants selected aroma and dilutions from our odor kits, they utilized our WebUI to test atomization performance and select desired smell release distance, intensity, and working mode to simulate the smell experiences based on their expected scenarios.
5. *Post-workshop Interview & Questionnaire (30 min):* We conducted a semi-structured interview and an online follow-up survey on satisfaction with the whole workshop and usability of the Olfackit. The participants described their experiences using our kits, such as the difficulty, encountered problems, and suggested features.

After this workshop finished, we rewarded each participant with a 5 mL perfume. Furthermore, the experimenters transcripted, processed and analyzed the data from the workshop, including the manuscript, voice, and video recordings.

8.2.1. Brainstorming and design themes

All participants rated the brainstorming method very high ($M = 4.60$, $SD = 0.66$), especially the prepared inspiration objects, demos, and additional prints (1 = strongly dissatisfied, 5 = strongly satisfied). During the brainstorming stage, the participants generated 108 smell-enhanced ideas, of which 90% were technically feasible to implement with OlfacKit.

To further understand their design intentions of using the sense of smell, we subdivided generated concepts based on (1) *target users*, (2) *usage scenarios*, (3) *integrated objects*, and (4) *design purposes*. The results show great diversity.

We found that the participants have created 27 different *usage scenarios* (e.g., playing video games, photographing, air-detection, watching the immersive performance, breathing training) for 18 types of *target users*, which includes both human(e.g., drivers, travelers, readers, barista, military) and non-human users (e.g., pets and plants). We also extracted 50 potential *physical forms* with which smell-enhanced technology can be merged. They ranged from wearables (e.g., hat, wristband, necklace), intelligent products (e.g., drone, laptop, sweeping machine, VR headset), and household items (e.g., garbage can, garment steamer, package box, curtains) to public space items (e.g., signs and exhibits). Additionally, we further summarized the *design purposes* into 14 categories, but not limited to: mental health and well-being (12), synesthesia design (9), cleaning and covering unpleasant odors(9), beauty and cosmetics(7), immersive experiences (6), flavor-augmentation (5), notification (5), branding (3), smell training (3), repellent (3), social communication (3), odor detecting (2), education and inspiration tool (2), and assisting technology for decision-making, recognition, tracking, camouflage, reward, and warning. We also found that some participants tend to strengthen their ideas by *highlighting the design of scented mist* (8), for example, strengthening the visibility by adding colorful light, modifying the shape of the burst, creating a foggy atmosphere, and" music fountain" effect. We attached the detailed results in the supplementary materials.

8.2.2. Implemented prototypes and design expectations

Our participants have implemented 13 prototypes (see Figure 14 and Table 1). We found that the solo designers tended to show personal preferences reflecting their daily routines (e.g., driving, playing Legos, or video games), while others were in groups because they shared similar design goals. Generally, the participants were positively satisfied with their implemented projects ($M = 4.50$, $SD = 0.50$) and believed the workflow helped and facilitated decision-making from rough ideas to solid construction ($M = 4.90$, $SD = 0.30$). They further rated their satisfaction on appearance ($M = 3.90$, $SD = 1.04$), attachments ($M = 4.0$, $SD = 1.0$),

Table 1. Along with Figure 14, this table reports the detailed design choices of the implemented projects, consisting 10 solo designs (P1, P2, P7, P8, P9, P10, P13, P14, P15) and 3 group works (P3 and P4, P5 and P6, P11 and P12). Unlike the others, P1 created two prototypes, with one designed for immersive in-car experiences (Figure 14(B)) and the other for his ongoing research (Figure 14(A)), integrating scents, lighting, and music to relieve anxiety for students.

Participant (Projects)	Time cost (min) model/assemble /test	Scent container	Atomizer diameter	Odors	Attachment/decoration	Odor dilution/ smell distance
P1(A)	20/28/15	Vertical	10 mm	Lotus, ocean, grass, filbert	Ring	3%/9–10 cm
P1(B)	40/20/14	Vertical	16 mm	Ocean, sunshine, osmanthus	Hook/3d models	5%/6–7 cm
P2(C)	25/12/6	Vertical	10 mm	Dirt, whisky	3d models	5%/12–14 cm
P3(D) P4(D)	16/19/4	Vertical	10 mm	Cinnamon	Ring /3d models	3%/12–14 cm
P5(H) P6(H)	6/20/18	Horizontal	10 mm	Sunshine, tea	none	5%/9–10 cm
P7(F)	24/12/7	Vertical	10 mm	Cypress	3d models	20%/12–14 cm
P8(I)	8/5/2	Vertical	16 mm	Coffee	Hair clip	20%/14–15 cm
P9(G)	49/8/9	Vertical	10 mm	Ocean	3d models	20%/9–10 cm
P10(L)	13/13/14	Horizontal	16 mm	Grass, coffee, ocean	Ring	5%/14–15 cm
P11(J) P12(J)	21/15/12	Vertical	16 mm	Litchi, milk	3d models	5%/20–21 cm
P13(M)	22/15/16	Vertical	16 mm	Grass, breeze, dirt, sunshine	None	3%/20–21 cm
P14(K)	8/15/7	Horizontal	16 mm	Tea	Hair clip	3%/20–21 cm
P15(E)	14/31/12	Horizontal	10 mm	Lemon, lavander	Cylinder-clip	5%/6–8 cm

Figure 14. The figure shows the implemented prototypes from the workshop, including A) an HCI research project for multi-modality design, B) smell-enhanced immersive in-car experiences for travel based on the changing location, C) an odor-augmented Batman, D) a meditation necklace guiding the breath, E) make-up mirror decorated with flower-like perfume containers, F) a shark warning sign by the sea using smelly odor, G) Lego Accessories for Sense of Smell, H) smell-inspired CMF for textile designers, I) smoke removal using coffee perfume, J) "Litchi Cat" releasing scents at checkout for branding, K) tea-flavored atomization for augmenting "*Three Pools Reflecting Moon*" as a notification, L) immersive gaming experiences when playing Zelda, M) scent carnival for building block music box.

and smell experiences controlled by WebUI ($M = 4.60$, $SD = 0.49$).

We found that several participants reported the outcomes were *beyond their expectations in terms of appearance, odors, attachment integration, and resolution of scent release*. For example, P4 mentioned: "*I have no experience in modeling, but the flower-like pendant looks more delicate and compact than I thought, and the black color matches my shirt a lot.*" P11 mentioned "*It is super adorable, and the scents (litchi and milk) and colors (red and white) also suit this cat!.*" P10 commented on the attachment: "*I thought it might be heavy to attach it to my console, but it fits well, I can barely feel it when I play the games, and the scent release direction is just right and even more flexible to adjust.*" Other participants commented on the scent control; P7 mentioned: "*I am surprised on the resolution of scent release distance and intensity controlled by WebUI*"; "*The alternating scent releasing makes the cat even more vivid*" (P11 and P12). Our participants also ***expressed their willingness to use the prototypes after workshop*** for long-term use (P1-project B, P2), decoration (P15, P4), emotion regulation (P11), iteration design (P7), and usability test of smell experiences in the wild (P1-project A, P7).

8.2.2.1. Implementing more creative ideas using OlfacKit.

All participants were positively satisfied with OlfacKit ($M = 4.30$, $SD = 0.46$) and agreed that OlfacKit could inspire them to generate more creative ideas ($M = 4.60$, $SD = 0.66$). Four participants expressed their interest and wish to use OlfacKit after the workshop. P7 wanted to extend his work (Figure 14(F) and mentioned that: "*I hope to design a series of wildlife warning signs, for example, the bear, that can be used in the jungle, creating an invisible barrier by releasing smelly odors that can drive away both human and bears.*" P10 mentioned: "*I'd like to build a digital daffodil releasing scents on Lego MOC (My Own Creation) and hope people can vote!.*" P7 shared more detailed ideas of using odors within the gaming scenarios: "*The scents can be integrated into the map and scenes that are not familiar in daily life, for example, the war games, RPGs (Role-playing Games) playing as a cat (game" Stay"), simulating an animal's sense of smell. I am very excited to try it out as research work.*"

8.2.3. Form design for olfactory interfaces

Overall, the participants expressed that the interactive design tool was simple to understand and could help save more time for modeling ($M = 4.40$, $SD = 0,80$), including the visual interface and workflow. We found that P1 and P9 spent more time than others because they built the external models after generating fabrication-ready designs (they rated their modeling skills as 5 points). We also found that the needed maximum odor number is four, and that might be the reason only linear (Figure 14(A, B, H, J, and L) and square (Figure 14(M) permutations were chosen.

8.2.3.1. Decision-making on scent container selection.

We found that the participants chose more vertical containers over horizontal ones, and we suspect that it is because the mechanism of the vertical type and the atomization direction is more straightforward to understand. Another reason might be that considering the assembly difficulty, we limited the combination of horizontal design with 3D models. Generally, nine participants, especially those who chose horizontal design (P5, P6, P10, P14, P15), mentioned their intentions were based on the *scenarios' physical space, device placement, scent release direction, and olfactory organ*, which matches our initial design considerations. Additionally, we also found the participants using the container for *form/shape imitation*. For example, in the situation without adding 3d models, the vertical container tends to be more invisible, while the horizontal can be attractive and expressive. P13 mentioned: "*The vertical type looks more straightforward and standard to merge with my Legos, and I hope* *the scent delivery devices will be invisible*", and P15 mentioned: "*The multi-channel design of the horizontal essentially looks like a flower, I do not have to import additional 3d models.*" Others used vertical containers combined with 3d models and mentioned: "*The two containers look like the barrels (weapon) of Batman and that is why I choose black resin to print*"(P2), "*We designed this shape because we wanted a rocket-like shape for this cat, with the scented mist acting like the jet*"(P11 and P12).

Additionally, six prototypes were designed by integrating with 3d models (Figure 14(B–J), of which two participants, P1 and P9, built the external models by themselves, while other participants downloaded free STL models from online websites (e.g., Thingiverse). We found that the participants tend to *add 3D models to strengthen design expression, social acceptance, and create new attachments for OIs*. P7 used the shark model with opening jaws indicating the hidden information of" Danger" and mentioned: "*I wanted to add red lights strengthening the mist in the mouth (like the blood) along with the horrible shark face to better convey my intentions.*" P9 created a new attachment method for Legos by measuring and calculating the size of Lego bricks, and he modified the appearance of both the container and PCB case. He mentioned: "*I hope the controllable scent release module can be assembled with Legos and further explored as Lego Accessories merging into Lego scenes.*"

P1, P2, P3, P4, P11, and P12 focused on social acceptance by *describing the prototypes as daily commercial products*. P11 and P12 chose a classic character from Japan, Lucky Cat, which is typically put on the front deck in the restaurant for good luck and fortune; they mentioned: "*Everyone likes Lucky Cat, and the scent-delivery device can be more natural and acceptable to use.*" P1 mentioned: "*I built a shell to make it more productized (project B), and chose black PLA to match my car interior.*" P3 mentioned: *I chose the daisy model to decorate and cover the container, like an actual necklace*; "*I would be more willing to use the device if it can be designed into a character I like (Batman)*" (P2). Interestingly, P1 added: "*while I think as a prototype for the research work (project A), it looks just right, and I use all-white design to avoid bias on scents.*" We also found that the participants tend to lower their score of satisfaction with the appearance when designing the container and PCB as a whole (Figure 14(B&I)). P1 and P8 advised further decreasing the control board's dimension. Some other participants tried to distribute the weight and dimension of the board by hiding it in the 3d model (Figure 14(J)) or by wearing it on the back of the neck (Figure 14(A&D)).

8.2.4. Odor selection

Odors are crucial parts when designing olfactory interfaces. We observed how the participants selected the target scents from our odor kits and further summarized their intentions: (1) Most participants chose a *concrete odor that reflects, simulates, or reproduces a natural scene or item from physical world*, such as the lotus, tea, dirt, and sunshine. Among these scents, ocean ($n = 4$), grass ($n = 4$), and sunshine ($n = 3$) were the most chosen fragrances, which we suspect were most commonly seen in real life and shown in video games and toys. (2) *Utilizing functional scents*. For example, P15 chose lemon for morning refresh and lavender for calming down at night; P8 utilized a coffee smell to cover the smoke, and P7 tried to use pine as an unpleasant odor to warn the travelers. (3) Some other participants *selected the odors based on the appearance of OIs*, such as shape and colors that are highly related to the characters and 3d models. P2 used the dirt and whisky representing Batman, and he mentioned: "*I hope the scents can match the spirit of Batman; for example, the dirt reminds me of the bat cave where Batman rebirthed. Moreover, this odor can also relate to the wet weather to be used as a weather forecast*"; P11 and P12 mentioned: "*We randomly chose red PLA for printing, but along with the white paint, we found the litchi and milk match it the most and also suits the restaurant scenario.*" (4) Some participants tend to *use alternative odors instead of abstract describing*. We found that not all participants could come up with a precise odor that was needed. They tried to describe it and looked into alternative solutions. P2 mentioned: "*I was looking for a scent for Justice. However, there was no such smell, so I chose dirt and wine to match Batman, while still pleasant to me.*" P7 tried to describe his needed odor as" *stink*", " *unpleasant*," "*smelly*" but not harmful to the human. He finally chose cypress essential oil because he personally disliked the odor.

Moreover, seven participants (P1, P2, P3, P4, P7, P8, and P14) reported that the odor kits inspired and facilitated their decision-making on idea construction. P8 reported that: "*I could not decide which odor to use until I smelt the coffee, which inspired and reminded me that people usually use coffee grounds to cover the smoke at Cafe.*" Some other participants gave more in-depth advice on the procedure of odor selection. P7 and P12 suggested using a Fragrance Blotter instead of smelling directly from the perfume bottles.

8.2.5. Assembly and interchangeable scenarios

Our participants considered the assembly process positively simple ($M = 4.40$, $SD = 0.80$) and did not require much hands-on ability (1 = strongly disagree, 5 = strongly agree). We found that the assembly time varied depending on the quality of printing items (need to polish using sandpaper or polisher if the printed parts do not fit well), the scent container's number, and appearance customization. Five participants (P3, P4, P11, P12, and P7) *improved the completeness of their final designs through additional hands-on craft*. Detailedly, P7 painted the" WARNING" with yellow color to highlight the content, while painted the conductive wires black to be invisible (Figure 14(F)); P3 and P4 designed a necklace chain using our jewelry findings (Figure 14(D)) and tested many ways of connecting the scent container to make it stable to be worn. Some participants (P5, P6, and P4) reported the difficulty of using the PCB switch and mentioned that it was hard to insert the conductive wires into the terminal block on the control board since they were too small to manipulate. P6, P7, P9, and P15 wanted to hide the conductive wires, while P1 expressed that the refilling method could be more convenient, for example, adding the liquid aroma from outside without opening the piezo holder.

Furthermore, we found seven participants *explored extended scenarios for their designs after assembly*. P9 tried more Lego bricks, joints, and hinges with his design and created a module that could change the scent release direction (Figure 14(G); P8, P11, and P12 envisioned a different scenario by changing the scent release direction. P11 added: "*if the scented mist is to the top (Figure 14(J), like wearing a bag, it can be set at the eating table releasing the odor of seasonal drink or juice to brand; and if the burst is to the bottom, like a rocket flying to the sky, we can locate it at the front table or checkout to farewell the customers.*" P10 tried to attach his design to the console and VR headset (Figure 14(L), and P15 explored two horizontal designs, trying to connect them through the channel using the PTFE tubes (Figure 14(E), she mentioned: "*I assembled it as the building blocks and it can be freer on the scent release direction.*" We also found that P3's work inspired P8 (Figure 14(D); he mentioned: "*I can imagine more locations the device can be worn and used on the body, such as the headband and brooch.*"

8.2.6. Rapid scent control and mist manipulation

The participants agreed that the WebUI workflow was easy to understand and manipulate ($M = 4.20$, $SD = 0.87$). Like the assembly time cost, it will cost more time if the participants choose more types of odors. Some participants (P7, P5, P6, P14) reported that the "Fast Test" was necessary and could quickly evaluate the assembled mechanism.

For the *desired scent release distance and intensity*, most participants chose the parameters based on their sensitivity to smell, the size of physical space, and the distance between the device and nose. More detailedly, the participants tended to *decrease the intensity when with additional airflow or need rapid smell switching*. For example, P1 mentioned: "*I hooked this device (project-B) on the car air conditioner, I could easily smell the scent with a mild burst*"; P5 and P6 mentioned: "*We hope a quick switching in between different scent scenarios for the textiles, so we decrease the dilution and smell distance.*" P7 and P8 *increased the intensity to cover unpleasant smells and repel target users* by selecting a large odor dilution (20%). P8 mentioned: "*I designed this device releasing the coffee scent directly to the ashtray and hoped it can be strong enough to cover the smoke*"; P7 add: "*Stronger intensity would be benefit the usability and effectiveness for repelling the travelers from the dangerous shark.*"

Moreover, some participants tended to *observe, design, and manipulate the visible burst*. Firstly, we found that during the fast test session, the participants observed the visible bursts for assisting the smell test. Secondly, five participants wanted to *highlight the mist* by increasing the quantity of atomization, adding light (P7), creating dynamic effects like the music fountain (P11, P12, and P13), and modifying the shape of the burst (P14). They decreased the odor dilution (3%), avoiding vigorous smell intensity, except for P7, using the scents in the wild. Especially, P14 chose a crafting work to cover the burst and release the scent through the hollow structure (Figure 14(K). Besides, we found that P1 and P10 tended to *hide the mist, avoiding visual distraction*. P1 mentioned: "*Under the scenario of driving, I prefer to smell the scents without any visible cue*," P9 mentioned that: "*It would be intriguing and increase curiosity to smell the ocean without any other hint for the users.*"

Other participants also expected and suggested WebUI involve commonly-used sensors for non-programmers, for instance, using the infrared sensor to trigger scents when people approached (P7, P8, P9), releasing different odors to adjust the rhythm of breathing during meditation with a heart rate sensor (P3 and P4); or have detailed timing setting for day and night refreshing (P15) and gaming(P7). P9, P7, and P12 further suggested improving the graphical design of WebUI instead of using describing wordings.

9. Discussion and key findings

9.1. Atomization-based olfactory interfaces

Through the workshop, our participants successfully generated numerous creative smell-enhanced ideas with detailed design categories, which could facilitate guiding, extending, and enlarging the ubiquity of using Olfactory Interfaces. Regarding aesthetics, we validated that novices with little modeling experience could also build complex, compact, and fashionable designs in color, shape, form, printing materials, and quality, while expert designers could efficiently create designs for commercial-grade products. We also verified that the implemented prototypes could functionally and naturally merge with existing objects (e.g., in-car air condition, ashtray, mirror, and console) or be designed as standalone items (e.g., necklace, Lego accessories, signs) under interchangeable scenarios. Nevertheless, more detailed design parameters should be considered for higher-level design goals. For example, we used black-and-red Dupont wires with terminal blocks for fast assembly and disassembly. In contrast, alternative solutions (conductive wires and connecting methods) can be tried, for instance, using the FFC (Flat and Flexible Cable), soft silicone-made wires, copper tapes, sewing textile-based conductive wires for fabrics, or trying magnets connecting circuits. Besides, implementing the PCB that controls a single odor as a second choice could potentially decrease the whole dimension of the final design if only one or two odors were chosen.

Additionally, the scented burst also plays an important role when designing atomization-based Olfactory Interfaces, and they can be designed into scenarios that naturally have water vapor, mist, or fog or used as shape analogy and metaphor. For example, we can merge it with the electric iron's spray, attach it to the mug with hot water, or act as waste air, jet, and fires for the toys (e.g., automobile, rocket, and cooking). It is worth mentioning that when a large quantity of atomization is devised (like Figure 14(M), we recommend selecting highly diluted odor sources utilizing less corrosive substrate liquid (e.g., water) and avoiding setting the device in a space with poor ventilation in case of lingering and cross-contamination (e.g., metal and plastic), especially, when covered with other items (like Figure 14(K). On the other hand, to use the scented mist more invisibly, we suggest lowering the burst height, decreasing the desired scent release intensity (through WebUI), or placing the device within a white or light environment. Besides, even though the participants did not use the" Cleaning Mode" of our WebUI during the workshop, we still consider it critical for long-term use to avoid clogging atomizers' micro-pores. Moreover, for a more convenient method of refilling, the present mechanism can be iterated by combining the silicone plug with a check valve to add the liquid aroma from the outside without opening the piezo holder. Further research should also draw on the atomization performance when using the OIs in the wild with wind.

9.2. Scents, 3D shapes, and olfactory imagery

Our interactive design tool supports combining the imported 3d model with the scent container, which enlarges and extends OIs through various physical forms. Regarding a concrete 3d shape or form with olfactory imagery, such as fruit, flower, grass, and dirt, we could operate these for enlightenment education (learning and exploring the physical world) involving young children through smell-enhanced inspiration tools (e.g., as P9's work in Figure 14(G) and skunk demo in Figure 12). Especially could showcase the objects and items that were not commonly and easily accessed within daily settings (e.g., Bamboo). Besides, using the modeling tool could also facilitate adding value for non-digital products and crafts (e.g., figures, resin models, IP design) to further increase user engagement and stickiness. Furthermore, when merging the containers with a concrete character (e.g., Batman and Lucky Cat), we recommend setting the scent containers and atomizers in social-acceptable locations by avoiding sensitive parts, such as the face, head, and eyes. Additionally, previous research has explored crossmodal correspondences between scents and two shapes (Hanson-Vaux et al., 2013; Metatla et al., 2019), called "Bouba" and "Kiki." We consider OlfacKit also promising for this research topic by exploring and understanding the design intention of how users integrate scents (e.g., type and number of odors) to a specific 3D shape (imported 3d model) in a designed manner.

9.3. Potentials of using odors

We offered a wide variety of scent materials and a precise scent control platform for the participants, which efficiently helped them simulate the smell experiences and accelerate the design iteration of Olfactory Interfaces. Through the workshop, we found more opportunities for using odors. Firstly, utilizing functional liquids (e.g., antibacterial aroma, liquid medicine, and hormones) would be essential and unique for atomization-based olfactory interfaces compared to other scent delivery mechanisms (e.g., heat, fan, and pump). More candidate liquids can be explored for both human and non-human olfactory systems, such as oxytocin, repellent, and water-based hydrosols. Besides, further experiments should address the atomization performance, dilution, and cleaning strategy for long-term use.

Secondly, the intention of using unpleasant odors positively draws new research questions. During the workshop, we found that the participants tended to use abstract and subjective words to describe their target odors, which caused a long-time communication and selection bias. Besides, we can't program scents alike colors and sounds but describe them using language and descriptors, which let researchers rethink convenient solutions for odor index within the HCI field. Hence, reflecting on the user's feedback and existing knowledge, we envisioned a digital scent database that enables and encourages users to share their smell experiences and descriptions, building a smell-driven experience-sharing community. It also supports keyword searching for the target odors through descriptors. Furthermore, adding and organizing the scent biography, such as the place of origin, history and story, function, and available odor source (e.g., brand and links), could also facilitate the users in targeting the desired odors and assist in idea construction and future smell-related interaction designs.

In addition, when testing and experimenting with unpleasant odors (Lu et al., 2020), we recommend selecting a short scent release distance and a lower smell intensity and conducting the smell control in a space with good ventilation in case of any uncomfortable influence on the people nearby.

9.4. Efficient kit for smell-enhanced research

Through the workshop, we found that OlfacKit has successfully supported the development of smell-enhanced research within various domains, for instance, the multi-sensory design within HCI (Figure 14(A), Wearables (Figure 14(D), Industrial and Product Design (Figure 14(B and G), immersive gaming experiences (Figure 14(L), as well as the Textile Design (Figure 14(H). We consider OlfacKit an efficient and promising design solution kit for multidisciplinary research facilitating iteration in design, fabrication, scent control, and interaction design. Further research can be conducted through collaborating domain-based in-depth knowledge, for example, implementing detailed connectors within the Digital Jewelry Design to utilize more jewelry findings (e.g., headpins, crimps, bails, and ear wires). We could also extend the collaborations by exploring smell-center experiences within Fine Art, such as immersive performance, script writing, new media art, digital photography, and filming.

9.5. Limitation and future work

In this work, we utilize the atomization-based scent delivery method as a starting point for creating a design toolkit for Olfactory Interfaces. Further research should also address different design criteria for alternative scent-releasing mechanisms (e.g., heat, fan, pump, valve, and air cannon). For instance, precise smell control, unique smell experiences, cleaning and refilling strategy for long-term use, and moreover, how users manipulate their associated sensations, such as warmth and vibration, for more creative designs or trade-offs applying alternative solutions.

Additionally, even though open-sourced code for most sensors is available online, some participants still wished that our kits involved input implements for non-programmers. Therefore, we plan to empower our WebUI by developing a built-in sensor API library through unified logic, assisting more novice users (e.g., students who majored in design and art) in building diverse interactions independently and efficiently. As suggested by P9, organizing and processing the sensors commonly used in daily life (e.g., infrared, lighting, and moisture sensors) can facilitate smell-enhanced toolkit design and explore a wide range of applications. In contrast, more specific input methods should be investigated for context-based applications, such as temperature, breathing, and heart rate sensors for smell-enhanced health monitoring.

Besides, further work should also focus on the detailed iteration of OlfacKit, for example, filleting the standard models to create smooth structures, exploring more attachment methods (e.g., screwing with existing connectors), as well as adding time-setting parameters for WebUI. More critical practical issues should be considered and tested in the wild, for instance, the relationship between the smell distance, intensity, and environment size.

10. Conclusion

In this paper, we present OlfacKit, a toolkit that enables novice users to create atomization-based olfactory interfaces via piezoelectricity rapidly. We showed how the interactive modeling tool facilitated the design process and creates diverse form factors in terms of scent container type, the capacity of the scent container, permutation of the multi-scent, arrangements, attachment, and the shape of containers. We demonstrated how the physical construction kits could enhance assembly, customization, and odor selection. We also showed how the WebUI helped the users without hardware programming skills to prototype the scent release on-demand. We further evaluated and validated our method's capabilities through six demonstrations and a user study. We hope our work could greatly benefit the emerging smell-enhanced applications and the toolkit construction for the sense of smell.

Disclosure statement

No potential conflict of interest was reported by the author(s).

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